The Quartic Force Law at Cumulant Order Four

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Abstract

Papers 14-20 establish a cubic information-geometric pattern: the force is governed by the third cumulant, the classical symmetry algebra is abelian (Cartan), higher-order classical extensions fail, and the Bures density-matrix lift restores the full nonabelian gauge while converting the jerk into a measurable leakage rate. The natural next question is whether this pattern survives at cumulant order four. The quartic force law should be generated by the fourth cumulant $T^{(4)}$, on an arithmetic skeleton with three distinct primes, $R = \mathbb{Z}/p_1p_2p_3\mathbb{Z}$. We sketch the expected structure and ask whether a new qualitative obstruction appears at order four that is absent at cubic order, beyond the representation-theoretic Cartan truncation already known. The bounded-support argument of Marcinkiewicz still guarantees that the integrability locus remains trivial, but the character of the obstruction may change qualitatively when three primes are present.

This paper poses the question, outlines the minimal new machinery, and carries out the first item of that machinery — the finite atlas on $\mathbb{Z}/30\mathbb{Z}$ (§5). The atlas resolves the *structural* half of the question. Order four is qualitatively new, and the new feature is sharp: read in the additive-character basis, the fourth-cumulant selection rule is the first to carry a **Gaussian (Wick-pairing) subtraction**, $\delta_{a+b+c+d\equiv 0}$ minus the three pairwise products $\delta_{a+b\equiv 0}\delta_{c+d\equiv 0}$. The cubic order has no such subtraction. The connected (genuinely quartic) part of the rule supports, on $\mathbb{Z}/30$, exactly 914 three-prime vertices with no quadratic or cubic shadow — the first non-pairwise prime interaction in the tower. The *dynamical* half — the higher-jet variational principle and the status of Ostrogradsky-type modes — remains open and is restated as such.

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1. The Pattern Established at Cubic Order

At cubic order the information-geometric mechanics yields: - Force law = cubic (Amari–Chentsov tensor $T = K_{CH}$, third cumulant). - Variational source = cubic action-energy theorem: the accumulated action $\mathcal{A}_3$ and instantaneous Noether energy $\mathcal{E}_3$ are distinct but linked by time translation (Paper 17). - Classical symmetry algebra = abelian Cartan $\mathfrak{u}(1)^2$ (Paper 14). - Higher-order classical extensions fail (Paper 17, Remark 3.2). - Quantum lift restores full $\mathfrak{su}(2) \oplus \mathfrak{su}(3)$ and converts the jerk into leakage (Papers 15, 18). - Long-time dynamics are recurrent and non-chaotic with leakage controlled by the jerk (Paper 19).

The pattern is cumulant order matched to arithmetic prime-axis complexity, with the classical obstruction abelian and the quantum completion nonabelian.

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2. The Fourth-Cumulant Case

For fourth-cumulant mechanics the natural arithmetic skeleton is a product of three distinct primes:

$$R = \mathbb{Z}/p_1p_2p_3\mathbb{Z} \cong \mathbb{Z}/p_1\mathbb{Z} \times \mathbb{Z}/p_2\mathbb{Z} \times \mathbb{Z}/p_3\mathbb{Z}.$$

The group algebra decomposes by CRT as $\mathbb{C}[R] \cong \mathbb{C}^{p_1} \otimes \mathbb{C}^{p_2} \otimes \mathbb{C}^{p_3}$. The associated categorical simplex is $Q - 1 = p_1p_2p_3 - 1$ dimensional.

The cumulant tower on this simplex predicts that the **fourth-order generalized force** lives at the jerk level (third derivative of η):

$$F_i^{(4)} := \frac{d^3\eta_i}{dt^3} = T^{(4)}_{ijkl}(v)v^jv^kv^l,$$

where $T^{(4)}$ is the fourth cumulant tensor. At this order the generalized force is the jerk, not the acceleration. The Nambu bracket becomes $(Q-1)$-ary with $Q-3$ macro-Hamiltonians. The classical symmetry analysis would again be governed by volume-preserving, CRT-tensor-compatible, real-cone-preserving derivations. The maximal such algebra is expected to be the Cartan subalgebra of $\mathfrak{su}(p_1) \oplus \mathfrak{su}(p_2) \oplus \mathfrak{su}(p_3)$, i.e., $\mathfrak{u}(1)^3$.

This naming has an immediate structural consequence: fourth-cumulant dynamics lives one derivative higher than the cubic case, so the Paper 17 cubic action-energy theorem cannot be copied by simply replacing T with $T^{(4)}$. One needs a genuine higher-jet variational principle, together with a clean distinction between accumulated action and instantaneous Noether energy. Whether such

a principle introduces Ostrogradsky-type modes, or whether the real-cone + CRT constraints kill those modes before they appear, is itself a new qualitative question at order four that does not arise at cubic order.

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3. The Open Question — New Qualitative Obstruction at Order Four?

Question. Does a new qualitative obstruction appear at fourth cumulant order that is absent at cubic order?

At cubic order the obstruction was purely representation-theoretic: nonabelian generators rotate the trivial character into nontrivial ones and exit the real positive cone. At fourth cumulant order the same mechanism is present, but three primes introduce new phenomena: - Triple conductor mixing (terms that entangle all three prime factors simultaneously). - Higher-depth wild ramification when any $p_i = 2$. - Possible new filtered extension classes Ext_{CH} of arity greater than 2.

It is conceivable that these new structures generate an obstruction that cannot be reduced to the cubic Cartan type — for example, a genuine filtered carry that survives even after all pairwise CRT factors are quotiented out.

Marcinkiewicz still closes integrability. The bounded-support argument that forces the categorical trace to be transcendental is alphabet-universal and applies unchanged at fourth cumulant order. Thus the integrability locus remains trivial. The question is whether the *character* of the obstruction changes qualitatively.

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4. Minimal New Machinery Required

To answer the question one needs: 1. The quotient-character adjacency algebra for three-prime CRT products (extension of Paper 11). 2. The fourth-cumulant Taylor expansion off the centroid (extension of Paper 10). 3. A quartic analogue of the Paper 17 action-energy theorem, including the higher-jet Noether energy and the status of possible Ostrogradsky-type modes. 4. The representation theory of volume-preserving derivations on the three-factor tensor product that also preserve the real positive cone. 5. An explicit finite atlas for at least one small three-prime row (e.g., $\mathbb{Z}/2 \cdot 3 \cdot 5\mathbb{Z} = 30$) to test whether new carry classes appear.

The 5-dimensional exact integrator of Paper 14 generalizes immediately to higher dimension and can be used for numerical checks once the atlas exists.

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5. Resolution: the order-four selection rule and the $\mathbb{Z}/30$ atlas

The minimal computation named in §4, item 5 — the finite atlas on $R = \mathbb{Z}/30\mathbb{Z}$ — is carried out here. It resolves the structural half of §3: a new qualitative feature *does* appear at fourth cumulant order, and it is the first appearance of a Gaussian (Wick) subtraction in the character-basis selection rule. The dynamical half (the higher-jet variational principle and the status of Ostrogradsky modes, §2) is untouched and remains open (§6).

5.1 The fourth-cumulant selection rule

Read the fourth cumulant $T^{(4)} = Q_{ijk\ell} = \partial_i\partial_j\partial_k\partial_\ell\Phi$ at the uniform point in the additive-character basis of $\mathbb{Z}/N$, exactly as the cubic Amari–Chentsov tensor is read in T3 (the conductor-lattice selection rule $a+b+c \equiv 0$). Contract $Q|_u$ against four non-trivial characters $(\chi_a, \chi_b, \chi_c, \chi_d)$. The moment–cumulant (connected-correlation) expansion, applied to the uniform categorical family — whose moments are *fully diagonal*, $\mathbb{E}[X_{i_1}\cdots X_{i_m}]|_u = N^{-1}\,[\,i_1 = \cdots = i_m\,]$ — gives

$$\boxed{Q(\chi_a,\chi_b,\chi_c,\chi_d)\big|_u \;=\; \delta_{a+b+c+d\equiv 0} \;-\; \delta_{a+b\equiv 0}\delta_{c+d\equiv 0} \;-\; \delta_{a+c\equiv 0}\delta_{b+d\equiv 0} \;-\; \delta_{a+d\equiv 0}\delta_{b+c\equiv 0}}$$

for $a, b, c, d \in \{1, \ldots, N-1\}$ (all congruences mod N). The rule is verified to machine precision on $N = 6$ and $N = 8$ by building $Q|_u$ from the partition formula and contracting against every non-trivial character quadruple (certificate `empirical/verify_quartic_selection_z30.py`). **[P]**

The qualitatively new feature. The cubic rule of T3 is the single connected term $\delta_{a+b+c\equiv 0}$, with *no* subtraction: the only non-trivial set-partition of three indices is $2+1$, which isolates a single character and vanishes on H_0 (where $\delta_{c\equiv 0} = 0$). At order four the $2+2$ partition survives — both blocks carry a non-trivial character — and the cumulant *subtracts* the three pairwise products. **Order four is therefore the first cumulant order whose selection rule carries a Gaussian (Wick-pairing) subtraction.** The fourth cumulant equals $+1$ exactly on the *connected* quadruples ($a+b+c+d \equiv 0$ with no vanishing sub-pair), 0 on the single-pairing quadruples (the Gaussian content cancels exactly), and is negative on the doubly-degenerate ones. The connected support is the genuine fourth-order content; the subtracted pairings are precisely the second-order (Gaussian) information that a quadratic instrument already carries. The new structure at order four is thus the **bifurcation of the selection rule into a connected (genuinely quartic) part and a disconnected (Gaussian, pairwise-reducible) part** — the first cumulant order at which the tower must separate "new" from "already seen at order two."

5.2 The atlas on $\mathbb{Z}/30$

On $R = \mathbb{Z}/30\mathbb{Z} = \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$, the conductor packets H_d ($d \mid 30$, $d > 1$; T3) are:

packet	$\dim = \varphi(d)$	primes of d	generators k
H_2	1	$\{2\}$	15
H_3	2	$\{3\}$	10, 20
H_5	4	$\{5\}$	6, 12, 18, 24
H_6	2	$\{2,3\}$	5, 25
H_{10}	4	$\{2,5\}$	3, 9, 21, 27
H_{15}	8	$\{3,5\}$	2, 4, 8, 14, 16, 22, 26, 28
H_{30}	8	$\{2,3,5\}$	1, 7, 11, 13, 17, 19, 23, 29

(Dimensions sum to $29 = 30-1 = \dim H_0$.) Of the 1202 unordered quadruples with $a+b+c+d \equiv 0$ (mod 30), exactly 120 are Wick-reducible (some sub-pair $\equiv 0$; the Gaussian floor) and 1082 are connected. Classifying the connected quadruples by the set of primes their four conductors span:

primes spanned	connected quadruples
$\{5\}$	4
$\{2,3\}$	6
$\{2,5\}$	32
$\{3,5\}$	126
$\{2,3,5\}$	**914**

The genuine three-prime obstruction anticipated in §2–§3 is now explicit and counted: 914 **connected quartic vertices couple all three primes simultaneously and are reducible neither to a Gaussian pairing nor to a pairwise (two-prime) face.** These are the order-four vertices with no cubic or quadratic shadow — the first non-pairwise content in the cumulant tower. **[P]**

5.3 What is resolved, and what remains

The atlas (item 5) is computed and the order-four selection rule is exact. The qualitative-newness question of §3 is answered for the cumulant-tensor structure: order four is qualitatively new, via the connected/Gaussian (Wick) bifurcation, and $\mathbb{Z}/30$ carries 914 genuine three-prime connected vertices. The selection rule says *where* the new content lives.

What remains is the *dynamical* content of §2–§4: the higher-jet variational principle (item 3), the status of Ostrogradsky-type modes, and the representation theory of volume-preserving cone-preserving derivations on the three-factor product (item 4) — i.e. whether the 914 connected three-prime vertices force a symmetry-algebra carry class beyond the cubic Cartan type. The selection rule locates the obstruction; whether it is a new dynamical carry is the next computation. On the *recovery* side (Paper 22's closed-leaf projection Π_τ onto the single-prime Cartan axes $A_2 \oplus A_3 \oplus A_5$), this carry is now settled to **vanish** at the finite order-four row: every one of the 914 cross-prime vertices contains a mixed-conductor character (none is single-prime-only, by a mod-2 parity argument using $\dim A_2 = 1$), so all lie in $\ker \Pi_\tau$ and carry no von Mangoldt weight — the finite §7 leakage-extinction instance (certificate `empirical/verify_leakage_extinction_z30.py`). What stays open is the *dynamical* carry (items 3–4 above) and the all-order limit (Paper 22, Conjecture 5.2).

6. Scope and Next Steps

The cumulant-order / prime-axis correspondence acquires a genuinely new feature at the three-prime row — the Gaussian-subtraction bifurcation of the selection rule (§5.1) — settled here at the level of the finite atlas (§5.2). This same connected/disconnected split is, in the language of the conductor-blind-spot program (Paper 34), the point at which the cumulant tower first distinguishes its genuine higher-order content from the second-order (Gaussian) floor; and in the Riemann-Hypothesis route of Paper 22 it is the order-four row $\mathcal{W}_\zeta(f * f^*)|_{\leq 4}$ on $\mathbb{Z}/30$, extending that paper's cubic recovery (its Section 5.3) to the first three-prime interaction.

The natural continuation is now the *dynamical* machinery of §4 (items 3–4): the higher-jet action-energy theorem and the representation theory deciding whether the connected three-prime vertices generate an obstruction beyond the cubic Cartan type. The 5-dimensional exact integrator of Paper 14 generalizes immediately and can be run on the atlas now that it exists.

Keywords: information geometry, Nambu-Fisher mechanics, quartic force law, fourth cumulant, three-prime CRT, Cartan obstruction, cumulant-order prime-axis correspondence.

Exploratory version — the question the series has not yet asked is now explicitly posed.